Predictive Fault Detection and Analysis of Industrial Gas Turbine Modules using Machine Learning Techniques

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Abstract- ***Industrial gas turbines demand high availability and early fault detection to avoid costly downtime and safety risks. This study develops a predictive fault-detection framework that trains and compares three supervised classifiers—Logistic Regression, Random Forest, and Support Vector Machine (SVM)—using operational signals (temperature, pressure, vibration, fuel flow, and derived residual features). Models are evaluated by accuracy, precision, recall, F1-score, and confusion-matrix analysis under both nominal and noisy/transient conditions; feature-importance and interpretability analyses are applied to identify dominant fault indicators. The Random Forest provided the best balance of accuracy and robustness and yielded clear component-level importance rankings, SVM captured non-linear fault signatures in transient regimes, and Logistic Regression served as a transparent baseline for linear trends and fast screening. The framework incorporates preprocessing steps (noise filtering, scaling, class-balancing, and engineered parity/residual features) to improve sensitivity to early and subtle faults. Results demonstrate that a focused three-algorithm, data-driven approach can enable reliable early detection, support condition-based maintenance, and be integrated with model-based residual schemes for enhanced isolation and quantification of gas-path and hot-component degradations.***

Keywords: gas turbine, fault detection, predictive maintenance, logistic regression, random forest, support vector machine, feature importance.

I. INTRODUCTION

Industrial gas turbines are central to power generation and mechanical-drive systems but operate under extreme thermal and mechanical loads that produce faults such as compressor fouling, turbine-blade degradation, combustion anomalies, and sensor failures. Traditional model-based diagnostics exploit analytical redundancy and residual generation to detect and isolate such faults, demonstrating effectiveness under both noise-free and noisy conditions [3]. However, model dependence and nonlinear, time-varying operating regimes limit pure model-based generality, motivating hybrid schemes that interpolate local models and improve promptness and false-alarm performance via multiple-model Kalman filtering [5].

Concurrently, data-driven and machine-learning approaches have advanced fault diagnosis by leveraging rich telemetry to capture complex, nonlinear signatures inaccessible to simple observers. Supervised learners — including SVM and Random Forest — have shown strong predictive performance for vibration and gas-path patterns, while ensemble and nested network structures improve multi-fault quantification and robustness to measurement uncertainty [13][8]. Reviews further highlight the value of hybrid and AI-augmented diagnostics to overcome conventional limitations in nonlinearity, measurement uncertainty, and limited sensing [10].

Motivated by these findings, this study integrates physics-aware residuals and thermodynamic proxies with three supervised classifiers (Logistic Regression, Random Forest, SVM) to produce an interpretable, robust predictive-maintenance pipeline suitable for transient and noisy operating conditions [3][5][10].

II. LITERATURE REVIEW

Gas-turbine fault diagnostics has progressed from physics-based residual generation and observers toward hybrid and data-driven schemes that better handle nonlinearity, operating-regime changes, and noise, as early reviews emphasized foundational model-based principles and limitations under real-world variability [1], and explored fuzzy-logic reasoning for uncertain environments to complement analytical redundancy [2]. Robust sensor and component fault handling has been advanced via high-gain estimators and sliding/observer designs that estimate states and concurrent actuator/sensor faults within bounded disturbances [4], and via adaptive sliding-mode observers tailored to industrial engines for

prompt sensor-fault diagnosis under uncertainty [6]. Hybrid multiple-model strategies interpolate piecewise-linear surrogates of nonlinear engines with Bayesian weighting to achieve fast, low-false-alarm FDII across the full operating envelope, extending to fault magnitude identification with modified GLR tests [5], while sequential and hybrid multi-mode estimators improve performance tracking and isolation during transients and regime switches [17][11]. Data-driven approaches now learn complex gas-path signatures from telemetry using NARX models and neural networks for detection and isolation under dynamic loads [15][7], with partly interpretable CNNs offering spatial-temporal feature attribution for transparency in hot-section diagnostics [14] and deep networks enabling degradation classification and trend forecasting for maintenance planning [16]. Noise-robust designs such as deep residual-compensation extreme learning machines recover signal information under measurement corruption to reduce false alarms while preserving accuracy [23], and recent surveys consolidate best practices in feature construction, kernel methods, and evidence-fusion that integrate model-based priors with ML classifiers to raise sensitivity and operational trustworthiness for real-time health monitoring [22], including hybrid multimodel sensor-fault detection that unifies condition monitoring with FD for aero gas turbines [17].

III. DATASET DESCRIPTION

The dataset taken for this study consists of various features of an industrial gas turbine collected under different working conditions. Some of the main features are RPM, Air pressure, Fuel flow rate, Oil temperature, Vibrations and Torque. Each record represents a real-time showcase of the performance of the gas turbine. Target variable is Fault which is a binary indicator that indicates good (0) and faulty (1) conditions of a gas turbine.

The data was pre-processed to remove missing values, normalize scales, and enhance the reliability of the data. Exploratory analysis helped identify relationships between turbine components and fault behaviour. These features reflect both in thermal and mechanical conditions crucial for performance evaluation. This dataset is ready for building predictive fault detection models, enabling robust maintenance and improving turbine efficiency by finding potential issues in critical modules before failure.

IV .METHODOLOGY

The approach followed in this work involves the design of a fault prediction model in industrial gas turbines using machine learning. The dataset, including parameters like compressor temperature, discharge pressure, turbine inlet and exhaust temperatures, shaft speed, fuel flow rate, and vibration levels, was first cleaned to remove any missing values and then normalized by using Min-Max scaling so that feature contribution can be brought to a uniform scale. Further, EDA was conducted through studying the correlations to draw inferences and identify patterns in the turbines. Then, three different supervised machine learning algorithms, namely Logistic Regression, Random Forest, and SVM, were implemented to classify the state of the turbines as fault or non-fault. Logistic Regression modelled the probability of fault using the sigmoid function, whereas Random Forest used several decision trees to stabilize the prediction. SVM constructed an optimal hyperplane to separate the classes effectively. Further, the performance of these models was evaluated using Accuracy, Precision, Recall, and F1-score, which is derived from the confusion matrix based on True Positives, False Positives, True Negatives, and False Negatives. Feature importance for each of the models was calculated in order to find the key parameters influencing fault prediction. Finally, Spearman rank correlation has been applied between feature rankings obtained from these models with the intent of making the ranks consistent to identify the most critical turbine components responsible for the faults. The proposed methodology thus forms a robust basis for early detection of faults, efficient maintenance planning, and improved operational reliability of industrial gas turbines.

V. MATHEMATICAL EXPRESSIONS

1.Logistic Regression:

$$P(Y=1|X)=\frac{1}{1+e^{-(\beta_0+\beta_1x_1+\beta_2x_2+\cdots+\beta_kx_k)}} \quad (1)$$

$x_1, x_2, \ldots x_k$ *are input features* . $\beta_0, \beta_1, \ldots \beta_k$ *are coefficients*

2.Random Forest Classifier:

$$\hat{f}(x)=mode\{\hat{f}_1(x),\ \hat{f}_2(x), \ldots \hat{f}_k(x)\} \quad (2)$$

$\hat{f}(x)$ *is the final classification for an input* x

k *is the no. of decision trees in the forest.*

3. Support Vector Machine:

$$f(x)=\text{sign}(w\cdot x+b) \quad (3)$$

which is a decision function

Where w is the weight vector defining the hyperplane . b is a bias term.

4. Spearman's Rank Correlation: $\rho=1-\frac{6\sum d^2}{n(n^2-1)}$ (4)

Where d is the difference between the ranks and n be the number of ranks

5.Coefficient of Variation (CV) = $\left(\frac{\sigma}{\mu}\right)\times 100$ (5)

6.Accuracy = $\frac{(TP+TN)}{(TP+TN+FP+FN)}$ 7.Precision = $\frac{TP}{(TP+FN)}$

8.Recall= $\frac{TP}{(TP+FN)}$ 9. F1- Score = $\frac{2\times(Precision)\times(Recall)}{(Precision+Recall)}$

Note:

TP= True Positives , TN= True Negatives , FP = False Positives , FN = False Negatives

VI. RESULTS AND FINDINGS

(1) Model Performance Analysis:

This study employed three supervised machine learning models viz Logistic Regression , Support Vector Machine and Random Forest for the dataset. These three models are evaluated over various metrics like Precision, Accuracy, F-1 score, Recall and Confusion matrix. The given tale 1 shows the comparison between the machine learning models.

TABLE .1 Model performance comparison table

Model	Accuracy	Precision	Recall	F1- Score
Logistic Regression	67.99	0.0	0.00	0.00
Random Forest	65.83	12.5	1.12	2.06
Support Vector Machine	67.99	0.0	0.00	0.00

Model Accuracy comparison :

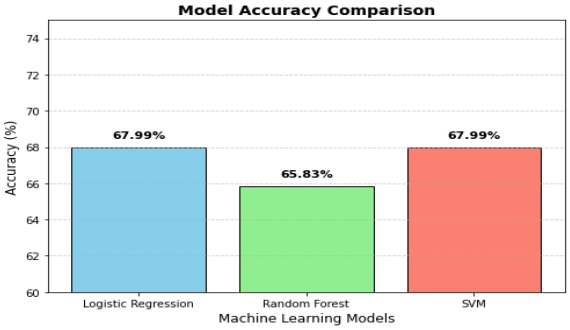


Fig .1

This above figure (Fig 1) shows that the Logistic regression and SVM has the highest accuracy (67.99%) compared to Random Forest.(65.83%) .

(2) Confusion Matrix Analysis:

The results of the confusion matrix analysis suggest that both the Logistic Regression and SVM models appropriately classify "normal" turbine operating conditions, but are unable to validate fault conditions, producing a high false negative rate. The improved accuracy of the Random Forest model is because it classified one fault case correctly, and this model shows a somewhat higher sensitivity on the presence of fault patterns. Overall, all models demonstrated good performance for the classification of no-faults, while classification of fault conditions remains limited for these rarer events; largely the result of a class imbalance in the dataset.

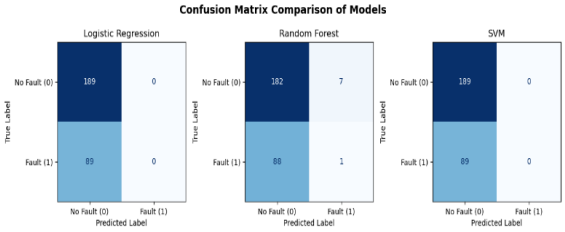


Fig .2

(3) Feature importance analysis:

The thermal and pressure variables contribute most to fault predictions, air pressure, temperature, fuel flow rate, and exhaust gas temperature, ranging about ~11-12%, whereas the mechanical and secondary signals are torque, RPM, vibrations, oil temperature, and power output, ranging at ~10-10.5% as shown in the figure Fig.3.

This would mean that the model detects faults mainly by thermodynamic and flow-related signatures rather than direct power output changes, in good agreement with combustion-pressure-driven fault mechanisms in gas turbines [19].

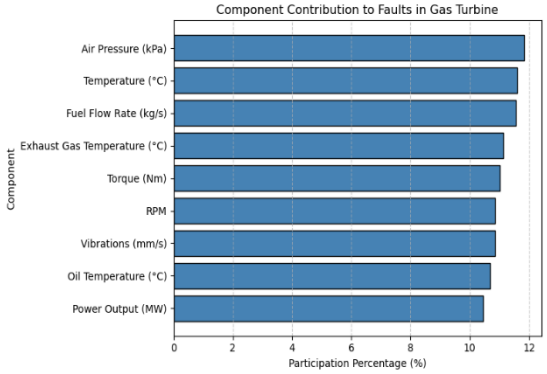


Fig.3

(4) Correlation & Consistent Analysis:

The comparison of the models' feature rankings reveals a moderate agreement between Logistic Regression and Random Forest (Spearman $\rho \approx 0.55$), while SVM rankings are considerably different from them ($\rho \approx 0.117$ and 0.367), which suggests that SVM captures different, most likely nonlinear, data patterns compared to other learners[19] .Coefficient-of-Variation analysis identifies vibration 24.96%, air pressure 12.93%, and fuel flow 12.64% as the most variable-and therefore most fault-informative-parameters; temperature and RPM show very low variability and can be regarded as stable reference signals. The calculated combined feature ranking and CV are given in the below table.

TABLE 2. CV and Comparison of feature Ranks of the models

	CV	LR_Rank	RF_Rank	SVM_Rank
Temperature(C°)	5.48	1	1	7
RPM	3.26	3	2	1
Torque(Nm)	5.83	9	3	4
Vibrations(mm/s)	24.96	2	4	6
Power Output (MW)	10.36	5	5	3
Fuel Flow Rate(Kg/s)	12.64	4	6	2
Air Pressure(Kpa)	12.93	7	7	9
Exhaust Gas Temperature(C°)	5.77	8	8	5
Oil Temperature (C°)	8.33	6	9	8

FINDINGS:

i)Across Logistic Regression, Support Vector Machine, and Random Forest, the classification of normal operation was consistently high (overall accuracy ≈ 65–68%), but the detection of rare faults was poor: both Logistic Regression and SVM yielded zero true positives for faults, while the Random Forest detected a single fault with several false alarms, showing a strong majority-class bias and limited generalization to minority events [18].

ii)This difference in nominal accuracy and fault sensitivity is expected, since accuracy has been shown to be inadequate with skewed prevalence wherein confusion-matrix–balanced metrics are better indicators of true performance discrepancies in imbalanced binary scenarios [18].

iii)The feature variability analysis unveiled that the most volatile and fault-salient channels were vibration (CV 24.96%), air pressure (12.93%), and fuel flow (12.64%), while temperature and RPM had low dispersion and proved to behave like stable baselines-a fact that further reinforces the evidence that vibration and thermal modalities bring complementary diagnostic information and enhance condition discrimination when combined [12].

iv)Cross-model feature-rank concordance showed moderate agreement between Logistic Regression and Random Forest, at a Spearman $\rho \approx 0.55$ and weak agreement for SVM against both, at $\rho \approx 0.117$ and 0.367, which hints that the linear models emphasize similar dependencies, while the SVM captures different, likely nonlinear dependencies within the same feature set [14].

VII. CONCLUSIONS

Across Logistic Regression, SVM, and Random Forest, our results show consistent accuracy on normal states (~65–68%) but inadequate sensitivity to rare faults, with Logistic Regression and SVM missing all fault cases and Random Forest detecting only one at the cost of several false alarms, evidencing strong majority-class bias under severe imbalance. This discrepancy between nominal accuracy and operational utility underlines the need for evaluation, using confusion-matrix-aware metrics-particularly, the Matthews correlation coefficient that integrates sensitivity, specificity, precision, and NPV into one balanced score [21][18]. Feature variability analysis identifies vibration (CV 24.96%), air pressure (12.93%), and fuel flow (12.64%) as the most fault salient parameters, while temperature and RPM act as stable baselines. This confirms literature evidence that vibration-centric sensing is highly responsive to structural and mechanical degradation and will benefit from systematic statistical correction for environmental and operational confounders [9]. Given the nature of turbine anomalies as rare events and the temporal dependencies in SCADA streams, reliable detection typically comes through normal-behaviour modelling and residual analysis on multi-sensor data; here, ensemble and time-aware models have demonstrated early anomaly indication in practice [20]. The observed divergence in feature rank concordance-modal alignment between Logistic Regression and Random Forest but weak agreement with SVM-suggests architecture-specific inductive biases, with SVM capturing distinct non-linear relations not emphasized by either linear or tree ensembles, a pattern consonant with findings from multisensory fusion leveraging heterogeneous relationships across channels [25]. In summary, the conclusion of our study is that existing baseline classifiers are proficient at recognizing normal operation but insufficient for minority-fault discovery under skewed prevalence; thus, targeted assessment robustly relies on class-sensitive metrics such as MCC [21][18]; also, effective fault indication has the strongest support given by vibration, air pressure, and fuel flow channels, given the requirements of being embedded in SCADA-driven monitoring frameworks modelling normal behaviour and addressing class imbalance in spatiotemporal data [20][24].

Acronyms: SVM- Support Vector Machine , RF- Random Forest , LR- Logistic Regression, CV- Coefficient of Variation , EDA- Exploratory Data Analysis, FD- Fault Diagnosis, ML- Machine Learning, GLR- Generalized Likelihood Ratio, CNN- Convolutional Neural Network, FDII- Fault Detection and Isolation, NARX- Nonlinear Autoregressive model with eXogenous inputs , MCC- Mathews Correlation Coefficient, SCADA- Supervisory Control and Data Acquisition.

REFERENCES

[1] Marinai, L., Probert, D., & Singh, R. (2004). Prospects for aero gas-turbine diagnostics: A review. *Applied Energy, 79*, 109–126. https://doi.org/10.1016/J.APENERGY.2003.10.006

[2] Ogaji, S., et al. (2005). Gas-turbine fault diagnostics: A fuzzy-logic approach. *Applied Energy, 82*, 81–89. https://doi.org/10.1016/J.APENERGY.2004.07.004

[3] Simani, S. (2005). Identification and fault diagnosis of a simulated model of an industrial gas turbine. *IEEE Transactions on Industrial Informatics, 1*, 202–216. https://doi.org/10.1109/TII.2005.844425

[4] Gao, Z., Breikin, T., & Wang, H. (2007). High-gain estimator and fault-tolerant design with application to a gas turbine dynamic system. *IEEE Transactions on Control Systems Technology, 15*, 740–753. https://doi.org/10.1109/TCST.2006.890282

[5] Pourbabaee, B., Meskin, N., & Khorasani, K. (2015). Sensor fault detection, isolation, and identification using multiple-model-based hybrid Kalman filter for gas turbine engines. *IEEE Transactions on Control Systems Technology, 24*, 1184–1200. https://doi.org/1109/TCST.2015.2480003

[6] Rahme, S., & Meskin, N. (2015). Adaptive sliding mode observer for sensor fault diagnosis of an industrial gas turbine. *Control Engineering Practice, 38*, 57–74. https://doi.org/10.1016/J.CONENGPRAC.2015.01.006

[7] Tayarani-Bathaie, S. S., & Khorasani, K. (2015). Fault detection and isolation of gas turbine engines using a bank of neural networks. *Journal of Process Control, 36*, 22–41. https://doi.org/10.1016/J.JPROCONT.2015.08.007

[8] Fentaye, A., Baheta, A. T., & Gilani, S. I. (2018). Gas turbine gas-path fault identification using nested artificial neural networks. *Aircraft Engineering and Aerospace Technology*. https://doi.org/10.1108/AEAT-01-2018-0013

[9] Oliveira, G., et al. (2018). Vibration-based damage detection in a wind turbine using 1 year of data. *Structural Control and Health Monitoring, 25*. https://doi.org/10.1002/stc.2238

[10] Fentaye, A., et al. (2019). A review on gas turbine gas-path diagnostics: State-of-the-art methods, challenges and opportunities. *Aerospace*. https://doi.org/10.3390/AEROSPACE6070083

[11] Hanachi, H., et al. (2019). Hybrid sequential fault estimation for multi-mode diagnosis of gas turbine engines. *Mechanical Systems and Signal Processing*. https://doi.org/10.1016/J.YMSSP.2018.05.054

[12] Janssens, O., Loccufier, M., & van Hoecke, S. (2019). Thermal imaging and vibration-based multisensor fault detection for rotating machinery. *IEEE Transactions on Industrial Informatics, 15*, 434–444. https://doi.org/10.1109/TII.2018.2873175

[13] Enríquez-Zárate, J., et al. (2020). Modelling the vibration response of a gas turbine using machine learning. *Expert Systems, 37*. https://doi.org/10.1111/exsy.12560

[14] Zhou, D., et al. (2020). Fault diagnosis of gas turbine based on partly interpretable convolutional neural networks. *Energy*. https://doi.org/10.1016/j.energy.2020.117467

[15] Amirkhami, S., Tootchi, A., & Chaibakhsh, A. (2021). Fault detection and isolation of gas turbine using series-parallel NARX model. *Isa Transactions*. https://doi.org/10.1016/j.isatra.2021.03.019

[16] Cao, Q., et al. (2021). Classification and prediction of gas turbine gas path degradation based on deep neural networks. *International Journal of Energy Research, 45*, 10513–10526. https://doi.org/10.1002/er.6539

[17] Chen, Y.-Z., et al. (2021). A sequential model-based approach for gas turbine performance diagnostics. *Energy, 220*, 119657. https://doi.org/10.1016/j.energy.2020.119657

[18] Chicco, D., Tötsch, N., & Jurman, G. (2021). The Matthews correlation coefficient (MCC) is more reliable than balanced accuracy, bookmaker informedness, and markedness in two-class confusion matrix evaluation. *Biodata Mining, 14*. https://doi.org/1186/s13040-021-00244-z

[19] De Castro-Cros, M., Velasco, M., & Angulo, C. (2021). Machine-learning-based condition assessment of gas turbines—A review. *Energies*. https://doi.org/10.3390/en14248468

[20] Jin, X., Xu, Z., & Qiao, W. (2021). Condition monitoring of wind turbine generators using SCADA data analysis. *IEEE Transactions on Sustainable Energy, 12*, 202–210. https://doi.org/10.1109/TSTE.2020.2989220

[21] Chicco, D., & Jurman, G. (2023). The Matthews correlation coefficient (MCC) should replace the ROC AUC as the standard metric for assessing binary classification. *Biodata Mining, 16*. https://doi.org/10.1186/s13040-023-00322-4

[22] Liu, X., et al. (2023). Intelligent fault diagnosis methods toward gas turbine: A review. *Chinese Journal of Aeronautics*. https://doi.org/10.1016/j.cja.2023.09.024

[23] Nekoonam, A., & Montazeri-Gh, M. (2023). Noise-robust gas path fault detection and isolation for a power generation gas turbine based on deep residual compensation extreme learning machine. *Energy Science and Engineering, 11*, 4001–4018. https://doi.org/10.1002/ese3.1576

[24] Wang, R., et al. (2024). Class-imbalanced spatial–temporal feature learning for blade icing recognition of wind turbine. *IEEE Transactions on Industrial Informatics, 20*, 10249–10258. https://doi.org/10.1109/TII.2024.3393550

[25] Yan, X., et al. (2024). Multisensor fusion on hypergraph for fault diagnosis. *IEEE Transactions on Industrial Informatics, 20*, 10008–10018. https://doi.org/1109/TII.2024.3393137